UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

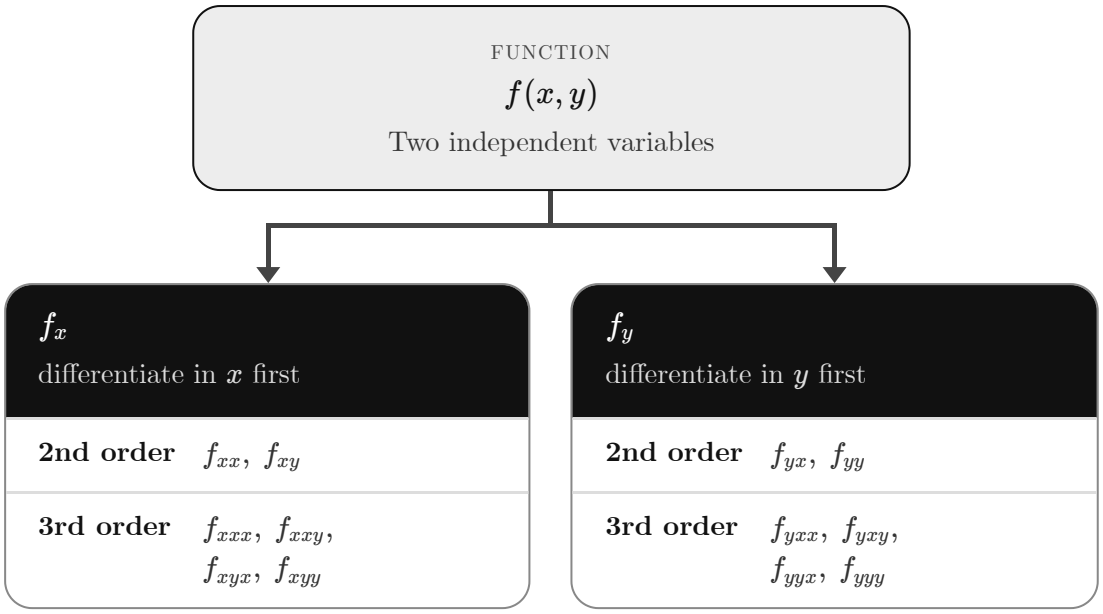
Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question

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3. Figure F4.2 · Which derivative answers which geometric question.
4. Figure F5.1 · Two slopes combine into the gradient.
5. Figure F6.1 · Reading concavity from a second-order partial.
6. Figure F7.1 · Verifying equal mixed partial derivatives.
7. Figure F7.2 · Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a).
8. One variable vs two variables

1 Higher-order partial derivatives

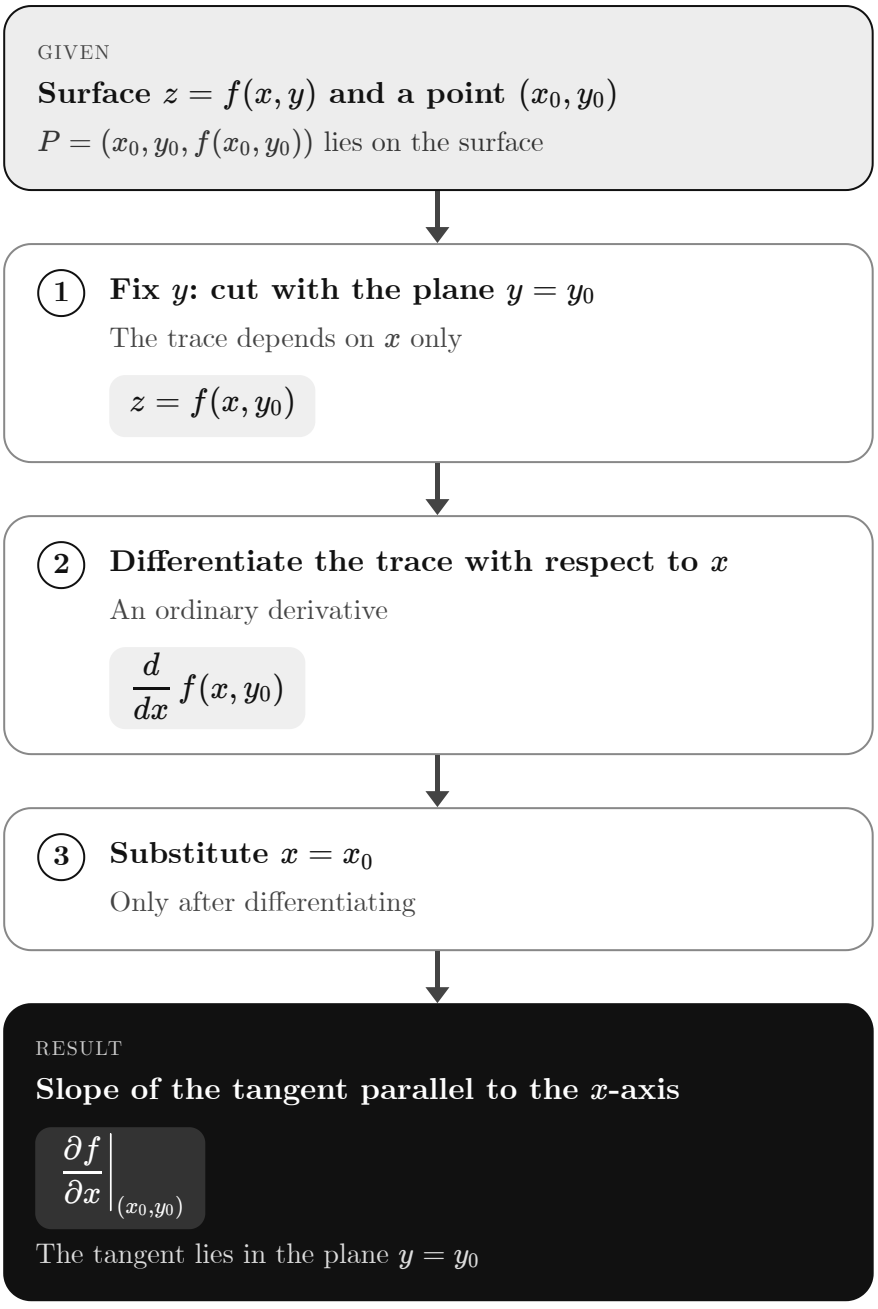


Remark. Totals: 2, 4, 8, ..., 2^n. With m variables there are m^n. When mixed partials are equal, only $n+1$ of the 2^n are different.

Figure F1.1. Counting partial derivatives of each order. ESA ★

S002·3 · T002 00:01:37 · T002 00:02:13

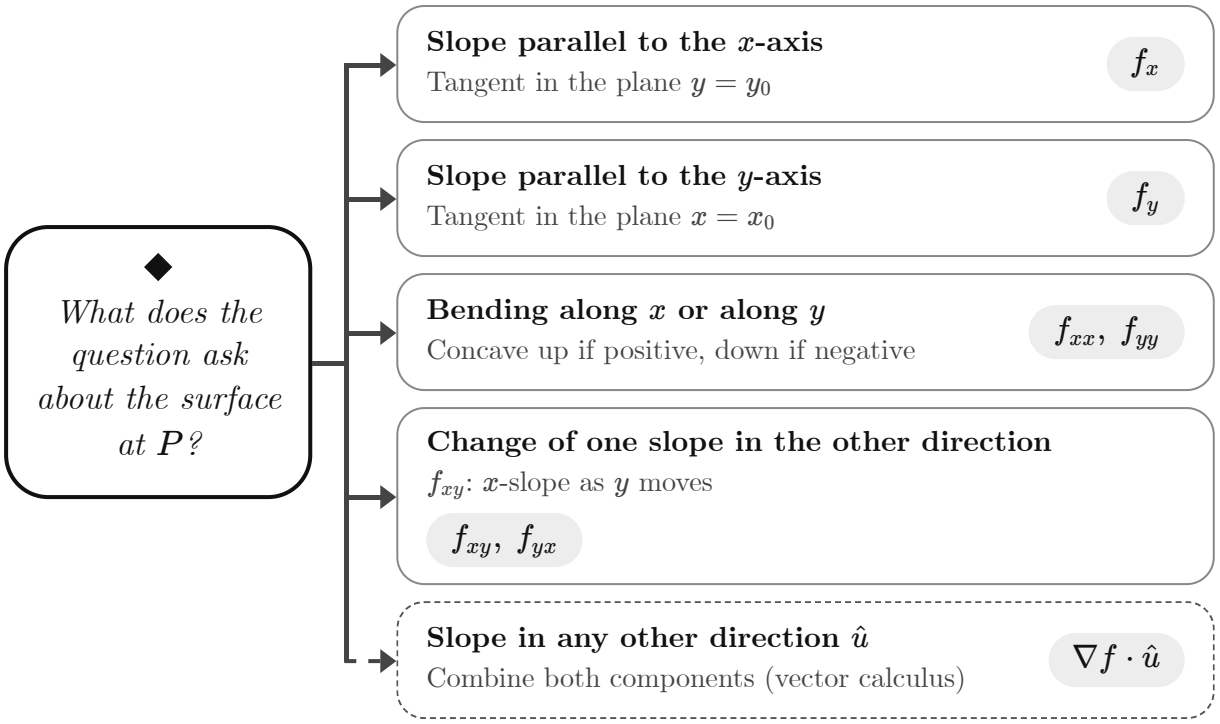
4 First-order partials as slopes of tangents



Remark. For $\partial f/\partial y$, swap the roles: cut with $x = x_0$, differentiate $f(x_0, y)$ in y, then put $y = y_0$.

Figure F4.1. Partial derivative as the slope of a tangent, step by step. ESA ★★

S002·6 · S002·7 · V002 00:07:45 · T002 00:06:10

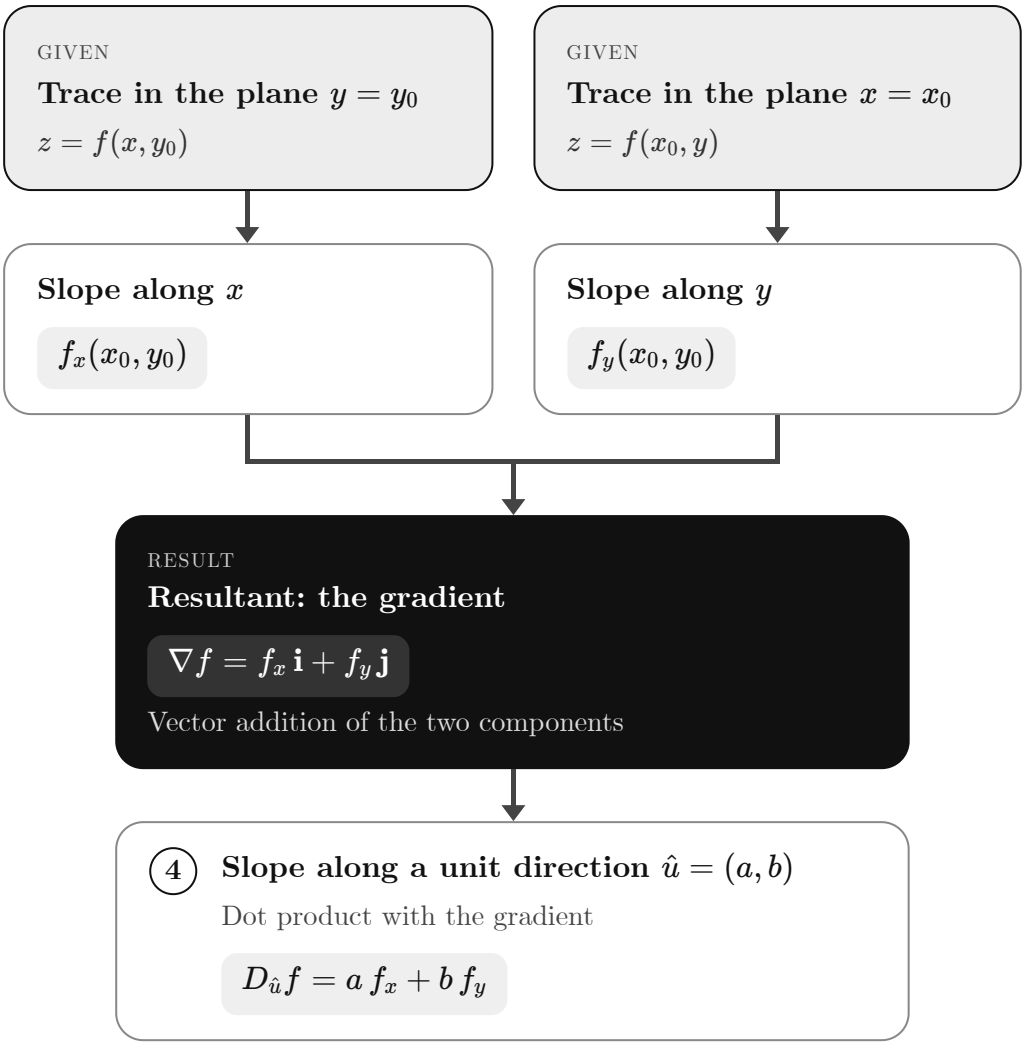


Remark. The dashed route is the gradient idea from slide 8. Its formula $\nabla f \cdot \hat{u}$ is added here for clarity and studied in vector calculus.

Figure F4.2. Which derivative answers which geometric question. ESA ★★

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5 Slope in any other direction: the gradient

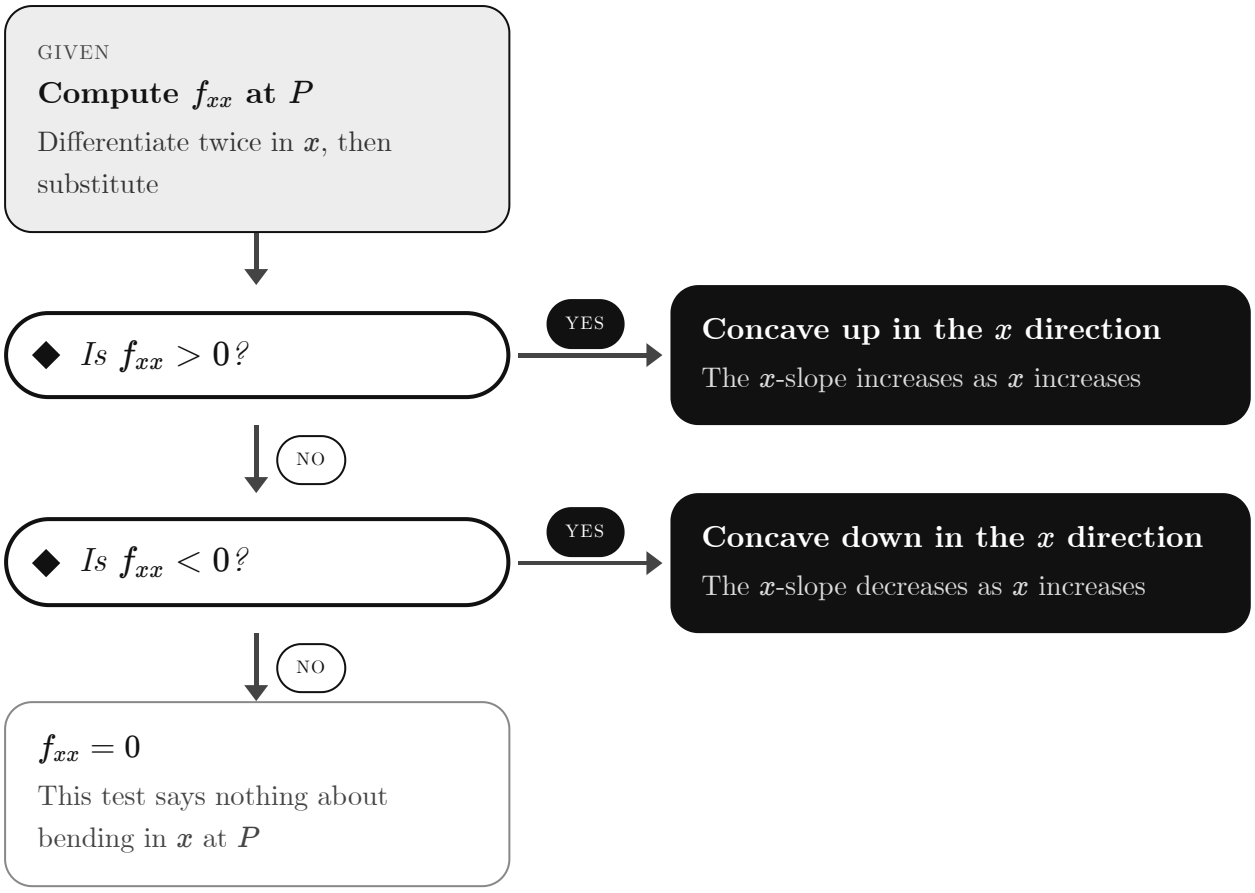


Remark. Check $\hat{u} = (1, 0)$: you get back f_x. The steepest slope is $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Example: the dome at $(1, -0.5)$ gives $\nabla f = (-2, 1)$ (W5.1).

Figure F5.1. Two slopes combine into the gradient. ESA ★

ADDED FOR CLARITY S002·8 · T002 00:09:06 · T002 00:09:23

6 Second-order partials: concavity and mixed partials



Remark. Run the same ladder with f_{yy} for the y direction. Read the sign of f_{xy} in the same way: positive means the x-slope increases as y increases (Figure G6.2).

Figure F6.1. Reading concavity from a second-order partial. ESA ★★

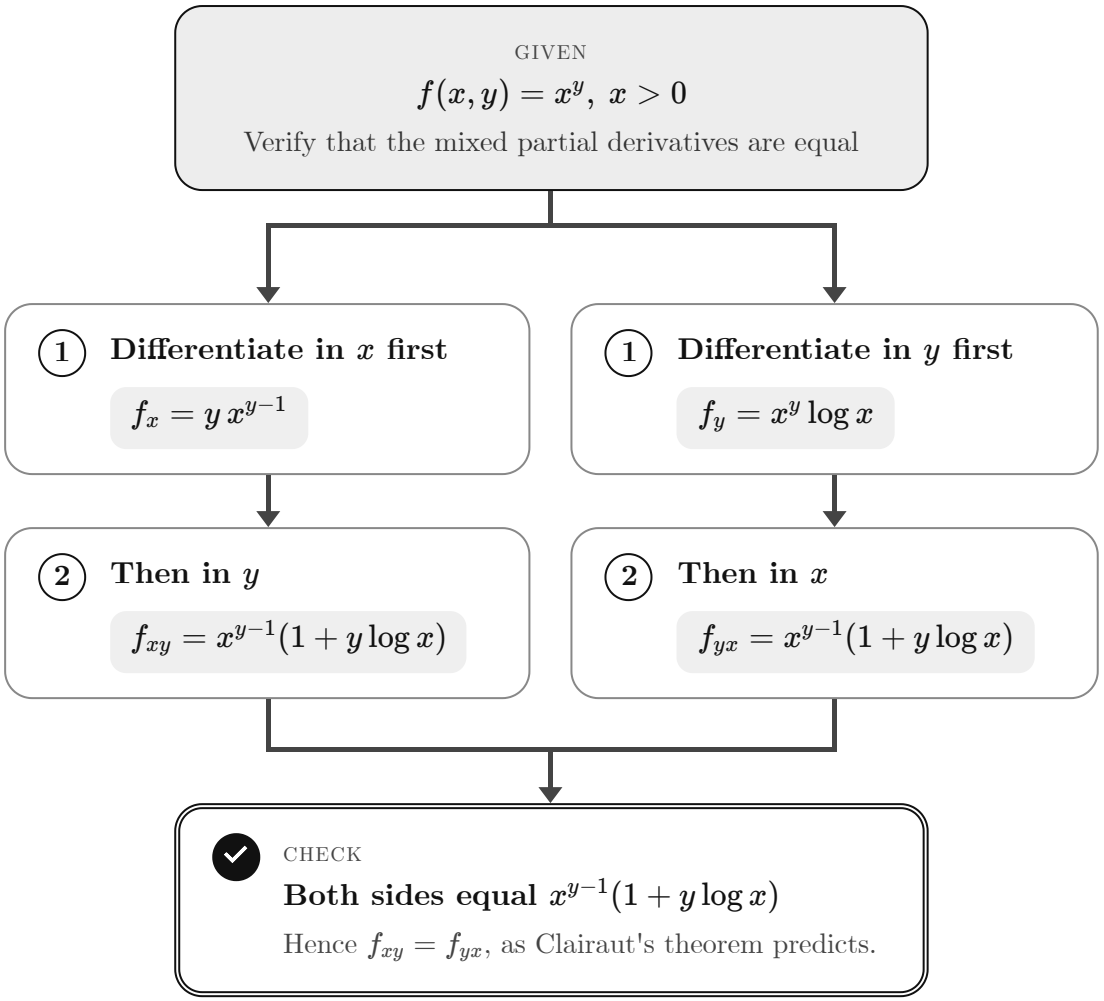
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One variable vs two variables

	One variable, $y = f(x)$	**Two variables, $z = f(x, y)$**
Graph	Curve in a plane	Surface in 3-D
Tangents at a point	One	Infinitely many, one per direction
First derivative	dy/dx: slope of the tangent	f_x, f_y: slopes parallel to the x- and y-axes; ∇f for other directions
Second derivative	d^2y/dx^2: concave up or down	f_{xx}, f_{yy}: concavity in x, y; f_{xy}, f_{yx}: change of one slope along the other direction
Number of order n	1	2^n (or m^n for m variables)

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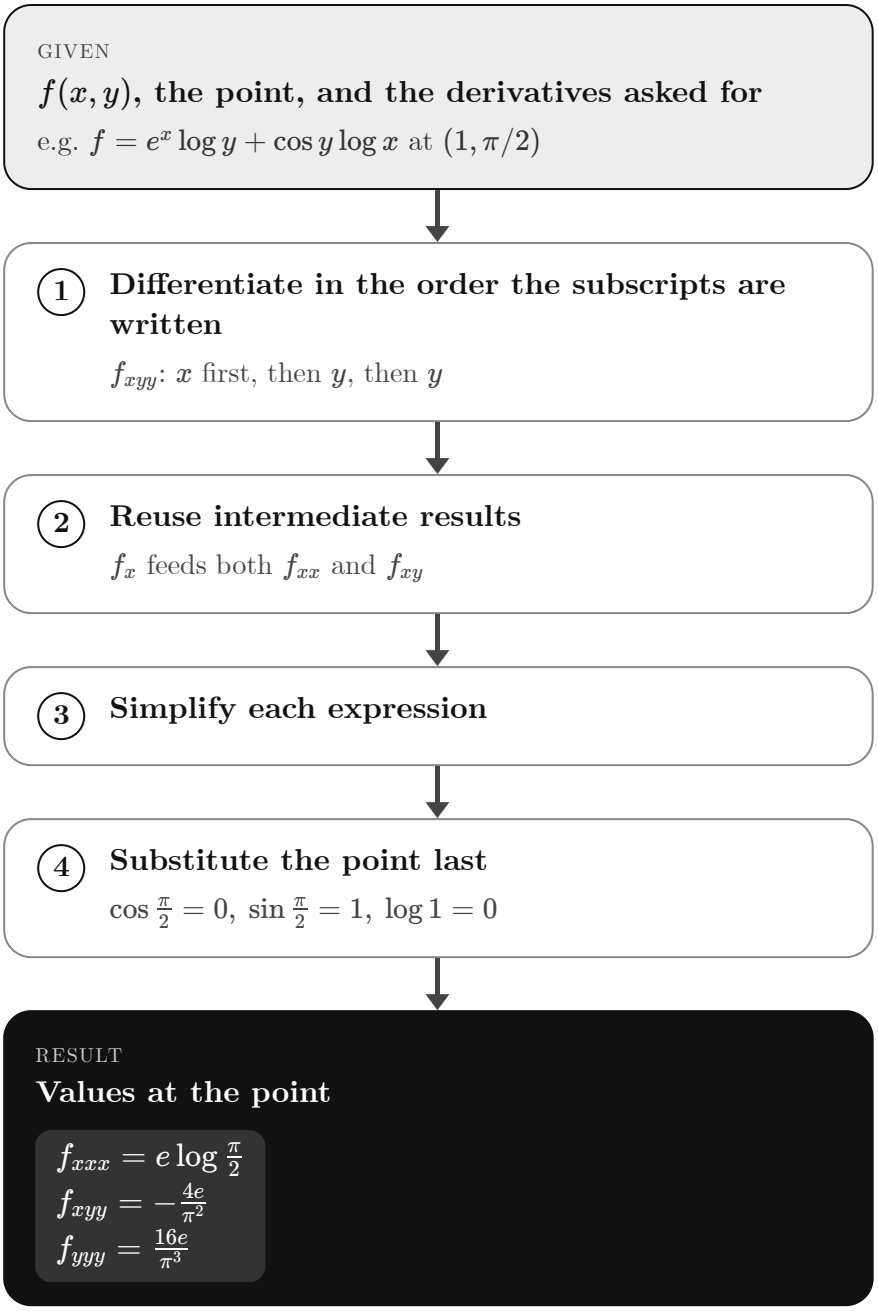
7 Problems on partial derivatives



Exam trap. In x, x^y is a power; in y, it is an exponential. Using the wrong rule is the usual slip.

Figure F7.1. Verifying equal mixed partial derivatives. ESA ★★

CW U1 P2 · T002 00:14:59



Remark. log is the natural logarithm, so $\frac{d}{dy}\log y = \frac{1}{y}$. Check one value against the CW answer key if you can.

Figure F7.2. Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a). ESA ★★

ESA Dec-23 1a · CW U1 P5 · S002·3